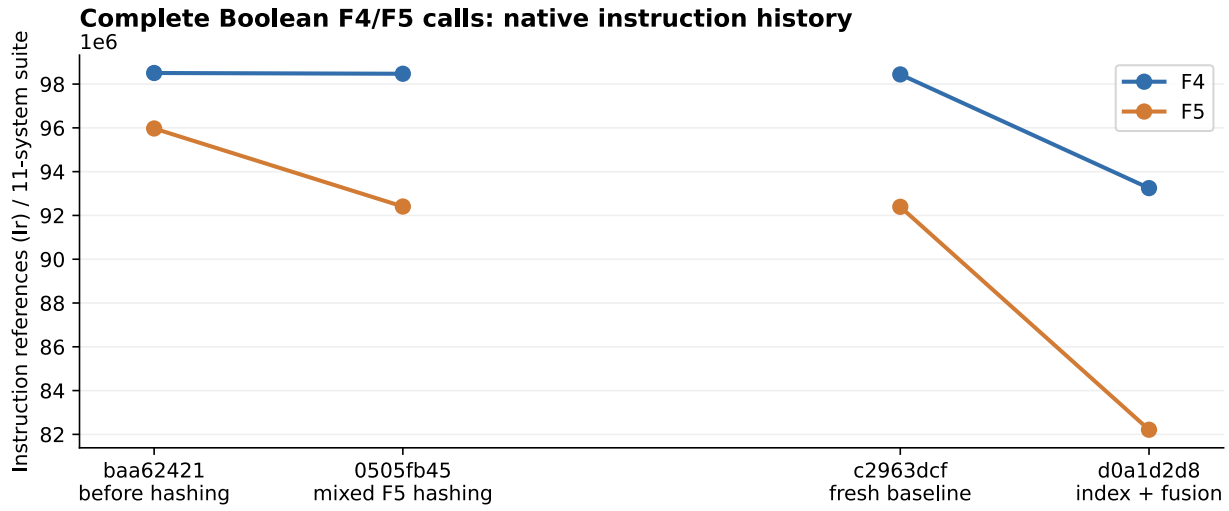
Complete Boolean F4/F5 calls

Native instruction and allocation measurements · run-20261012T0543Z



Each segment: 5 paired rounds, 11 systems, native medians and all samples; merged workers.
run-20261012T0543Z · Gap crosses unrelated main updates; both studies' comparison.json.

F4: 98,440,694 → 93,248,010 Ir per suite. Paired instruction reduction 5.28% (95% bootstrap interval 5.274% to 5.284%). Native verdict: fewer instructions.

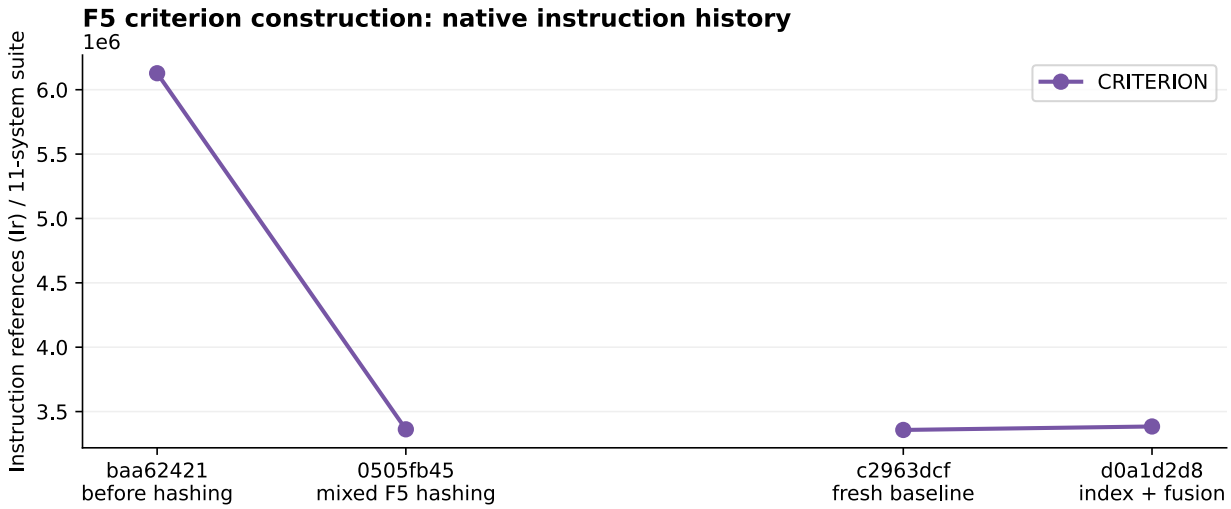
F5: 92,393,939 → 82,211,306 Ir per suite. Paired instruction reduction 11.02% (95% bootstrap interval 11.016% to 11.032%). Native verdict: fewer instructions.

Five paired rounds of all eleven frozen systems, one Rayon worker. Both complete-call arms are optimized. The native verdict requires a 95% bootstrap interval excluding zero and an effect above twice the A/A noise estimate and 2%. Every reduced output, input hash, pruning report and word-XOR count matched. All complete-call cases have the fewer-instruction verdict. The clarified complete-call gate passes; the original gate over 33 cases fails on three standalone criterion cases. ACCEPTANCE-CLARIFICATION records this scope change before final measurement.

Native sources: final comparison and raw-record hashes, frozen protocol, first rejection, bounded correction, gate scope, fixture repair, report and validation. Raw Callgrind dumps and Rust JSONL remain beside the data. Earlier segment: hashing study. Validation: 3,995 release-library tests passed, zero failed, 112 ignored; complete named coverage verified. The long ignored isogeny test passed separately. 46 Groebner/F4, 18 F5, 21 polynomial, 24 native coordinate and five comparison tests passed; Cargo examples and Clippy passed. Earlier failed/interrupted checks are retained. The inherited fixture-only repair preserves order rejection and strict parsing. Runtime and process RSS remain unmeasured.

F5 criterion construction

Native instruction and allocation measurements · run-20261012T0543Z



Each segment: 5 paired rounds, 11 systems, native medians and all samples; merged workers.
run-20261012T0543Z · Gap crosses unrelated main updates; both studies' comparison.json.

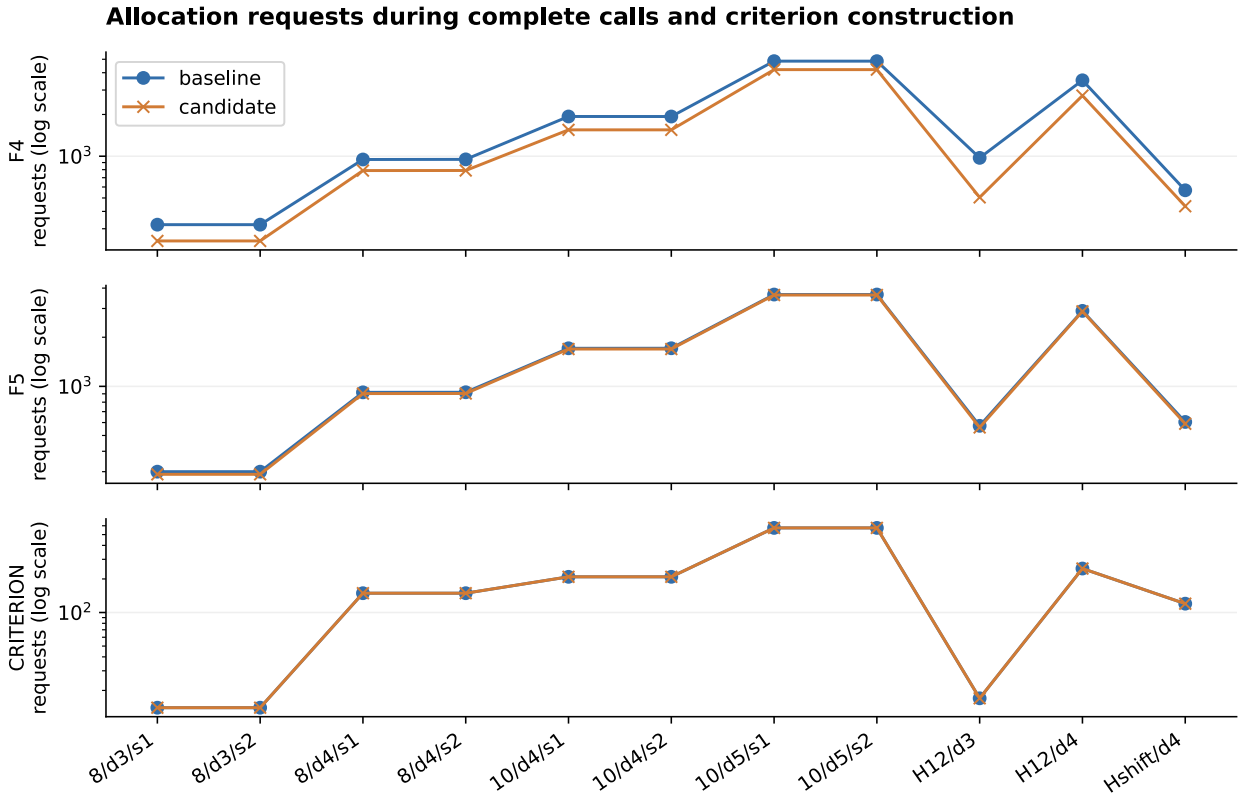
CRITERION: 3,357,250 → 3,384,189 Ir per suite. Paired instruction reduction −0.80% (95% bootstrap interval −0.808% to −0.791%). Native verdict: neutral.

Criterion production source/rules are unchanged; the standalone aggregate is neutral. Three cases have more-instruction verdicts: the two n8/d3 seeds (reductions −2.31% and −5.16%) and holdout n12/d4 (−4.07%). Their cause is not established here, and the original all-arm gate fails. F5 counts full F4 rows and builds selected rows in one traversal; F4 uses a memory-bounded direct index and sorted readback terms. Pruning rules, signature order, row limits and counted XOR work are unchanged. Standalone criterion cost is a phase result; it is already included in complete F5 calls and is not added a second time.

Native sources: final comparison and raw-record hashes, frozen protocol, first rejection, bounded correction, gate scope, fixture repair, report and validation. Raw Callgrind dumps and Rust JSONL remain beside the data. Earlier segment: hashing study. Validation: 3,995 release-library tests passed, zero failed, 112 ignored; complete named coverage verified. The long ignored isogeny test passed separately. 46 Groebner/F4, 18 F5, 21 polynomial, 24 native coordinate and five comparison tests passed; Cargo examples and Clippy passed. Earlier failed/interrupted checks are retained. The inherited fixture-only repair preserves order rejection and strict parsing. Runtime and process RSS remain unmeasured.

Allocation requests

Native instruction and allocation measurements · run-20261012T0543Z



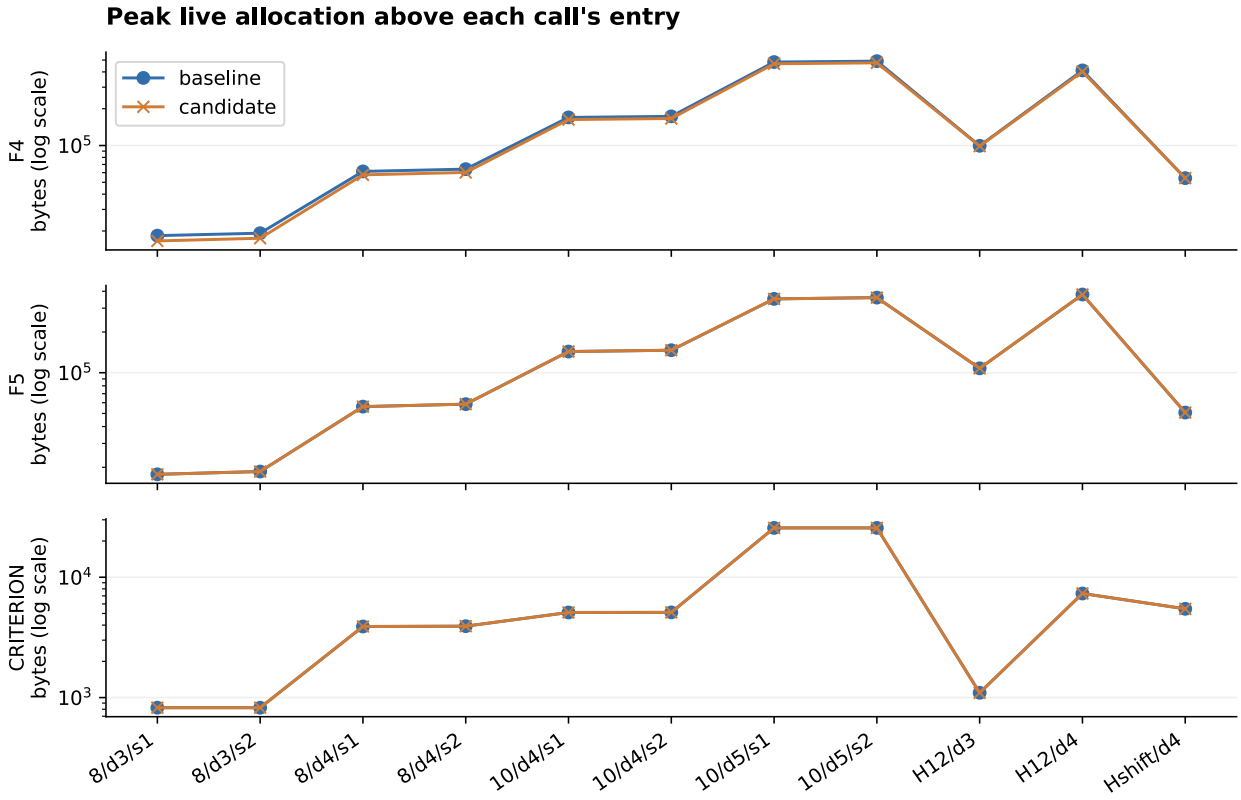
run-20261012T0543Z · Median of 3 native primary calls per system and variant.
H = frozen holdout; Hshift uses bits 52–63 and 6 multiplier variables.
Allocator requests; not elapsed time. Source: records/attempt-2-fixed/comparison.json.

The graph displays the native median allocation-request counts for each workload and variant, over three primary repetitions. F4 uses a bounded compact index and avoids a second term vector during readback. F5 avoids the separate full-row counting pass. Both complete-call arms include the unchanged packing and elimination work. Raw primary records retain requests and cumulative requested bytes separately. No memory metric regresses on any input in primary or paired controls.

Native sources: final comparison and raw-record hashes, frozen protocol, first rejection, bounded correction, gate scope, fixture repair, report and validation. Raw Callgrind dumps and Rust JSONL remain beside the data. Earlier segment: hashing study. Validation: 3,995 release-library tests passed, zero failed, 112 ignored; complete named coverage verified. The long ignored isogeny test passed separately. 46 Groebner/F4, 18 F5, 21 polynomial, 24 native coordinate and five comparison tests passed; Cargo examples and Clippy passed. Earlier failed/interrupted checks are retained. The inherited fixture-only repair preserves order rejection and strict parsing. Runtime and process RSS remain unmeasured.

Peak live allocation bytes

Native instruction and allocation measurements · run-20261012T0543Z



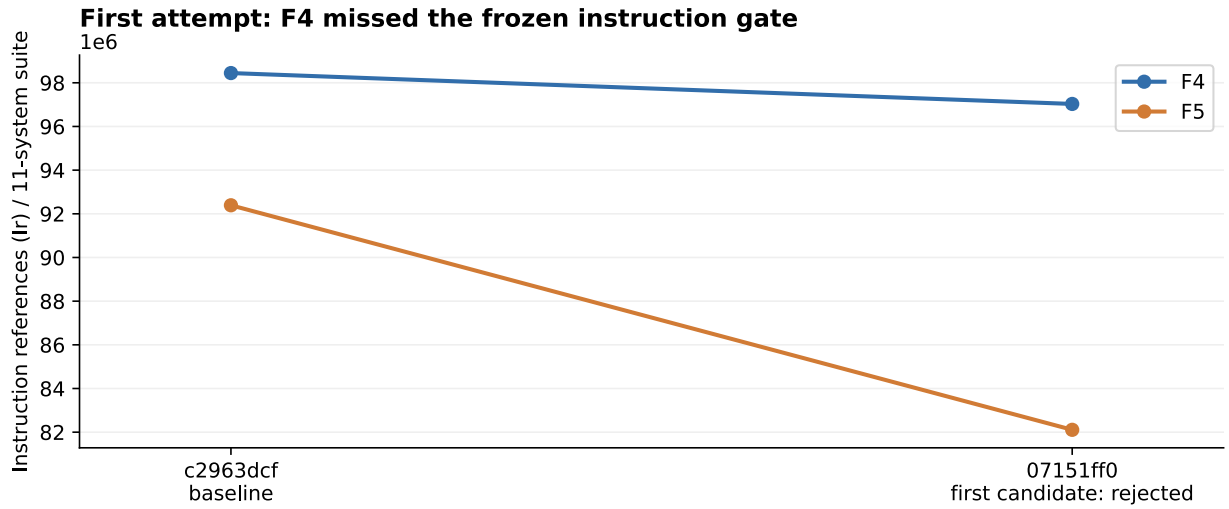
run-20261012T0543Z · Median of 3 native primary calls per system and variant.
H = frozen holdout; Hshift uses bits 52–63 and 6 multiplier variables.
Same-interval peak minus entry; not RSS or cumulative traffic. Source: records/attempt-2-fixed/comparison.json.

Native primary records retain three repetitions per variant, all eleven inputs, live bytes at entry and the absolute live-byte peak. The graph shows native medians of peak minus the entry value over that same allocation interval. Entry records separate retained inputs and warm outputs from the call’s additional peak. No allocation-request, requested-byte, absolute-peak or additional-live-byte regression occurred in primary or paired controls. The complete reduced outputs and retained warm inputs are identical, and the criterion allocation control is unchanged. Process RSS and elapsed runtime were not measured; the host lacks the pressure data needed for controlled timing.

Native sources: final comparison and raw-record hashes, frozen protocol, first rejection, bounded correction, gate scope, fixture repair, report and validation. Raw Callgrind dumps and Rust JSONL remain beside the data. Earlier segment: hashing study. Validation: 3,995 release-library tests passed, zero failed, 112 ignored; complete named coverage verified. The long ignored isogeny test passed separately. 46 Groebner/F4, 18 F5, 21 polynomial, 24 native coordinate and five comparison tests passed; Cargo examples and Clippy passed. Earlier failed/interrupted checks are retained. The inherited fixture-only repair preserves order rejection and strict parsing. Runtime and process RSS remain unmeasured.

Rejected first attempt

Native instruction and allocation measurements · run-20261012T0543Z



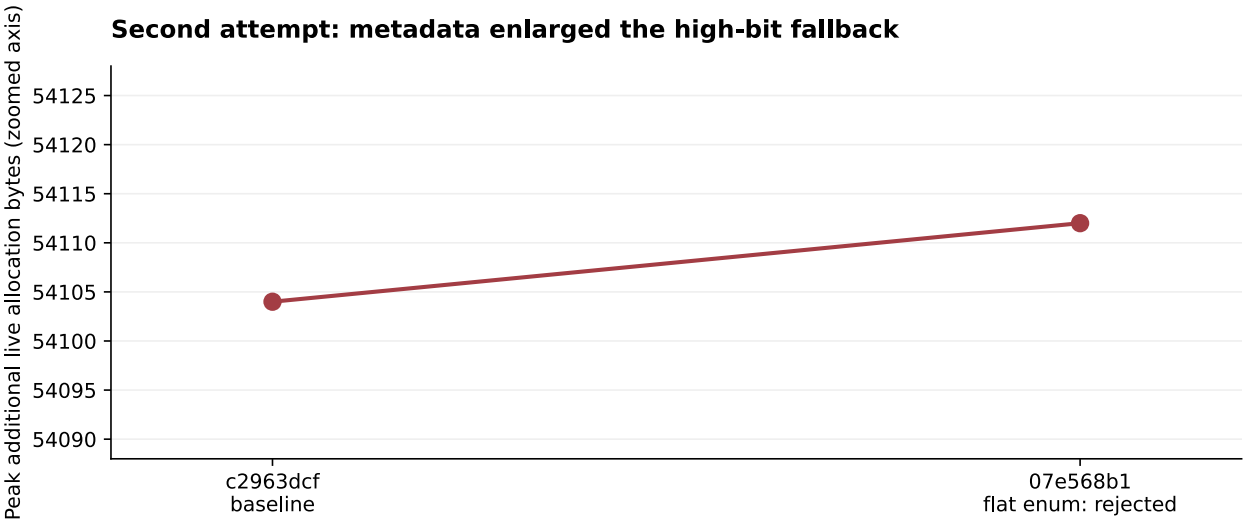
run-20261012T0543Z · 5 paired rounds. F4 verdict neutral; two-arm acceptance failed.
All first-attempt records retained. Sources: records/comparison.json and AMENDMENT-2.md.

The first candidate combined ordered F4 readback with default fused F5 construction. The native comparison records 1.44% fewer F4 instructions, below the frozen 2% verdict threshold, so F4 is neutral. F5 recorded 11.13% fewer instructions; outputs/accounting matched and memory did not regress. The frozen two-arm acceptance failed. All profiles, the comparison and controls are retained unchanged. AMENDMENT-2 preregistered compact F4 column lookup before the second measurement, preserving inputs and gates. The final attempt reuses the two unchanged baseline profiles and runs fresh candidate primary and paired profiles.

Native sources: final comparison and raw-record hashes, frozen protocol, first rejection, bounded correction, gate scope, fixture repair, report and validation. Raw Callgrind dumps and Rust JSONL remain beside the data. Earlier segment: hashing study. Validation: 3,995 release-library tests passed, zero failed, 112 ignored; complete named coverage verified. The long ignored isogeny test passed separately. 46 Groebner/F4, 18 F5, 21 polynomial, 24 native coordinate and five comparison tests passed; Cargo examples and Clippy passed. Earlier failed/interrupted checks are retained. The inherited fixture-only repair preserves order rejection and strict parsing. Runtime and process RSS remain unmeasured.

Rejected metadata representation

Native instruction and allocation measurements · run-20261012T0543Z



run-20261012T0543Z · Median of 3 primary high-bit F4 calls; confirmed in all 5 pairs.
Memory gate failed. Sources: records/attempt-2/comparison.json and native layout-probe.json.

The second implementation recorded 5.51% fewer F4 instructions and 11.03% fewer F5 instructions, but failed the memory gate. The high-bit fallback's additional peak grew from 54,104 to 54,112 bytes across all primary and paired calls. The native probe measured the original index metadata at 48 bytes, the flat three-way enum at 56 bytes, and a nested representation at 48 bytes. The heap-resident column layout accounts for the observed eight-byte increase. METADATA-FIX explicitly extended the two-attempt stop rule by one bounded correction and rerun. It preserved inputs and gates, added a metadata-size regression assertion and retained all failed results. The superseded full-library run was interrupted, not counted as passed.

Native sources: final comparison and raw-record hashes, frozen protocol, first rejection, bounded correction, gate scope, fixture repair, report and validation. Raw Callgrind dumps and Rust JSONL remain beside the data. Earlier segment: hashing study. Validation: 3,995 release-library tests passed, zero failed, 112 ignored; complete named coverage verified. The long ignored isogeny test passed separately. 46 Groebner/F4, 18 F5, 21 polynomial, 24 native coordinate and five comparison tests passed; Cargo examples and Clippy passed. Earlier failed/interrupted checks are retained. The inherited fixture-only repair preserves order rejection and strict parsing. Runtime and process RSS remain unmeasured.